# 1. Matrix Differentiation Applied in Linear Regression

$Y = X\beta +$

$L^{'}L = \ '$

$L^{'}L = (Y - X\beta)^{'}(Y - X\beta)$

$L^{'}L = (Y^{'} - \beta^{'}X^{'})(Y - X\beta)$

$L^{'}L = Y^{'}Y - Y^{'}X\beta - \beta^{'}X^{'}Y + \beta^{'}X^{'}X\beta$

$L^{'}L = Y^{'}Y - 2\beta^{'}X^{'}Y + \beta^{'}X^{'}X\beta = A + B$

$$\frac{d(L^{'}L)}{d\beta} = -2X^{'}Y + 2X^{'}X\beta$$

$f(\beta) = \beta^{'}X^{'}Y$

$f(\beta + d\beta) = (\beta + d\beta)^{'}X^{'}Y$

$f(\beta + d\beta) - f(\beta)$ **=** $\beta^{'}X^{'}Y + d\beta^{'}X^{'}Y - \beta^{'}X^{'}Y$

$f(\beta + d\beta) - f(\beta)$ **=** $d\beta^{'}X^{'}Y$

$f(\beta + d\beta) - f(\beta) = Y^{'}Xd\beta$ as **(**$d\beta^{'}X^{'}Y) = scalar = (d\beta^{'}X^{'}Y)'$

$$\left[\frac{df(\beta)}{d\beta}\right]^{'} d\beta = Y^{'}X\,d\beta$$

$$\frac{df(\beta)}{d\beta} = X^{'}Y$$

$F(\beta) = \beta^{'}X^{'}X\beta$

$$F(\beta+d\beta)=(\beta+d\beta)' X' X(\beta+d\beta)$$

By definiton

$$\left(\frac{dF}{d\beta}\right)' d\beta=[F(\beta+d\beta)-F(\beta)]$$

$$\left(\frac{dF}{d\beta}\right)' d\beta=(\beta+d\beta)' X' X(\beta+d\beta)-\beta' X' X\beta$$

**Let X'X be α**

$$\left(\frac{dF}{d\beta}\right)' d\beta=(\beta+d\beta)'\alpha(\beta+d\beta)-\beta'\alpha\beta$$

$$d\beta \;\; itself \;\; is \;\; small \;\wedge\; d\beta'\alpha d\beta \;\; is \;\; even \;\; smaller \;\wedge\; neglected$$

$$\left(\frac{dF}{d\beta}\right)' d\beta=(\beta'\alpha+d\beta'\alpha)(\beta+d\beta)-\beta'\alpha\beta$$

$$\left(\frac{dF}{d\beta}\right)' d\beta=\beta'\alpha\beta+\beta'\alpha d\beta+d\beta'\alpha\beta-\beta'\alpha\beta$$

$$\beta'\alpha d\beta=d\beta'\alpha\beta$$

Both terms are scalar

$$\left(\frac{dF}{d\beta}\right)' d\beta=2\beta'\alpha d\beta=2\beta'(X' X)d\beta$$

$$\left(\frac{dF}{d\beta}\right)'=2\beta'(X' X)$$

$$\left(\frac{dF}{d\beta}\right)=2(X' X)'\beta \qquad \left(\frac{dF}{d\beta}\right)=2(X' X)\beta$$